Leniency IV

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- Consider linear instrumental variables (IV) model when number of instruments K large. Arises in 2 ways:
    1. Interact low-dim instrument with controls (Angrist and Krueger 1991; Belloni et al. 2012).
    2. Use fixed effects (group indicators) as instruments: leniency design.
- Leniency designs brought to economics by Kling (2006). Exploits quasi-random assignment of decision-makers (judges, radiologists, ambulance workers, patent examiners, child welfare investigators) who differ in their propensity to grant treatment—*leniency*.

Key issues

1. monotonicity
2. two-stage least squares (TSLS) biased with many instruments. Alternatives?
3. standard errors
4. How does other stuff change? (First stage-F, complier characteristics, etc)

Setup and Identification

Estimation

Inference

Other considerations

Summary and illustration

- Same notation as in previous lecture, with reduced form and first stage

$$Y_i = Z_i'\delta + W_i'\psi_Y + u_{Yi}, \tag{1}$$

$$D_i = Z_i'\pi + W_i'\psi_D + u_{Di}. \tag{2}$$

  Parameter of interest is $\beta := \frac{E[\pi'\tilde{Z}_i Y_i]}{E[\pi'\tilde{Z}_i D_i]}$, where $\tilde{Z}_i = Z_i - E[Z_i W_i']E[W_i W_i']^{-1}W_i$.
- $X_i = (Z_i', W_i')'$ collects the right-hand side (RHS) (or "exogenous") variables.
- Nothing specific to leniecy IV! Only that $K = \dim(Z_i)$ may be large, and potentially also $L = \dim(W_i)$.

- For any matrix A, let $H_A = A(A'A)^{-1}A'$ denote hat matrix (also called a projection matrix), and let $\ddot{A} = A - H_W A$ denote residuals after projecting A onto the covariates.
  - $\tilde{Z}_i$ is population analog of $\ddot{Z}_i$
- Call $\ell_i = z'\pi + W_i'\psi_D$ (absolute) leniency, and call $\ddot{\ell}_i = \ddot{Z}_i'\pi$ relative leniency: propensity to grant treatment of decision-maker assigned to i *relative* to other decision-makers who could have been assigned to i

Simple judge design

- Individual i assigned judge J_i, random conditional on i's geographic location G_i.
- Covariates and instruments both indicators: $W_{i\ell} = \mathbb{1}\{G_i = \ell\}$, and $Z_{ik} = \mathbb{1}\{J_i = k\}$.
- There are L locations and $K + L$ judges; in each location we drop one judge to avoid collinearity—call this the reference judge.
- $\psi_{D,\ell}$ is average sentencing rate of the reference judge in location ℓ, and π_k is sentencing rate of judge k relative to reference judge in same location.
- Then ℓ_i is average sentencing rate of judge assigned to i, while $\ddot{\ell}_i$ is their leniency relative to average sentencing rate in location G_i.

Causal interpretation of TSLS estimand

- General theorem from last lecture specialized to setup where (i) Z_i is discrete and (ii) $\tilde{Z}_i'\pi = E[D_i \mid Z_i, W_i] - E[D_i \mid W_i]$ (relative leniency correctly specified) gives Lemma 2 in Goldsmith-Pinkham, Hull, and Kolesár (2025):

Lemma

Under (i) and (ii), $\beta = \frac{E[\lambda_i(Y_i(1)-Y_i(0))]}{E[\lambda_i]}$, where

$$\lambda_i = \sum_{k>j} \Pr(Z_i = z_k \mid W_i) \Pr(Z_i = z_j \mid W_i)(z_k'\pi - z_j'\pi)(D_i(z_k) - D_i(z_j))$$

- Under Imbens and Angrist (1994) monotonicity $z_k'\pi > z_j'\pi \implies D_i(z_k) > D_i(z_j)$ so $\lambda_j \geq 0$.

Monotonicity i

- Monotonicity assumption requires that judges agree on ranking of defendants, they just disagree on cutoff at which they start sentencing them.
- At odds with economic theory as well as statistical evidence:
    1. Chan, Gentzkow, and Yu (2022) point out that decisions of physicians differ due to both skill and preferences.
    2. Mueller-Smith (2015) argues that a judge may not use a common cutoff for each defendant: for instance, the cutoff may vary by crime type.
    3. Kleinberg et al. (2018) find that the increase in crime associated with judges who are more likely to release defendants on bail is about the same as if these more lenient judges randomly picked the extra defendants to release on bail.
    4. Statistical tests of monotonicity reject in empirical applications (e.g. Frandsen, Lefgren, and Leslie 2023; Agan, Doleac, and Harvey 2023; Coulibaly et al. 2024)

Monotonicity ii

5. Direct tests of monotonicity using judge panels also reject (Sigstad 2026)

- Is this a problem?
    - Evidence from Sigstad (2026) suggests problem not first order.

Average monotonicity

- Frandsen, Lefgren, and Leslie (2023) suggest a subtle fix: assume away negative weighting problem by assuming $\lambda_i \geq 0$. Called average monotonicity.
    - If Z_i is a single binary instrument, average monotonicity same as Imbens and Angrist (1994) monotonicity.
- Interpretation? With many instruments, many complier groups: one for each pair of decision-makers. Individuals can be compliers for some pairs, defiers for other pairs. From previous lemma, we know $\lambda_i \geq 0$ if no individual defier "on average".
- In particular, can write $\lambda_i = \Pr(D_i = 1 \mid W_i)\Pr(D_i = 0 \mid W_i)(\bar{\ell}(1) - \bar{\ell}(0))$, where $\bar{\ell}_i(1)$ is a (weighted) average absolute leniency of those who would release i, and $\bar{\ell}_i(0)$ that of those who would detain i.

Setup and Identification

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Other considerations

Summary and illustration

- Consider asymptotics in which $K = \dim(Z_i)$ and $L = \dim(W_i)$ can grow with sample size
- Denote population partial R^2 from adding instruments to first-stage regression by $R^2 = \sum_i \ddot{\ell}_i^2 / (\sum_i \ddot{\ell}_i^2 + n \operatorname{var}(u_{Di}))$, and denote expectation (under homoskedasticity) of first-stage F-statistic by

$$E[F] = \frac{\sum_i \ddot{\ell}_i^2}{K \operatorname{var}(u_{Di})} + 1$$

- If $L/n \to 0$, then ordinary least squares (OLS) bias $\to (1 - R^2) \operatorname{cov}(u_{Y,i} - u_{D,i}\beta, u_{D,i}) / \operatorname{var}(u_{D,i})$

Paper	F	n	K	L	R^2
Agan, Doleac, and Harvey (2023) Table III(3)	2.6	67,060	314	1,260	0.01
Farre-Mensa, Hegde, and Ljungqvist (2020)	1.6	32,514	4,238	2,401	0.07
Angrist and Krueger (1991), Table V(6)	4.7	329,509	30	21	0.00

Lemma

The TSLS estimator suffers from own observation bias towards OLS. In particular, consistency of TSLS in general requires $E[F] \to \infty$. Under homoskedastic errors, its asymptotic (relative to OLS) bias is given by $\frac{1}{(1-R^2)E[F]}$.

TSLS bias scales with F, rather than K/n! Rule of thumb $F > 10$ works!

- Bias arises because TSLS relative leniency $\hat{\ell}_{\text{TSLS},i} = \ddot{Z}_i'\hat{\pi}$ used by TSLS puts positive weight on own treatment status D_i.
- Total net weight, holding overall instrument strength constant, scales linearly with K, so TSLS bias scales with number of instruments.

Simple judge design

Without covariates, $\hat{\beta}_{\text{TSLS}} = \sum_i Y_i\hat{\ell}_{\text{TSLS},i}/\sum_i D_i\hat{\ell}_{\text{TSLS},i}$, and $\hat{\ell}_{\text{TSLS},i}$ is average sentencing rate of judge i is assigned to—including i's own treatment. If # cases per judge same, puts weight K/n on D_i.

Early solutions

- Solution 1 (Bekker 1994): use limited information maximum likelihood (LIML), or variants thereof. Only need $\sqrt{K}E[F] \to \infty$ for consistency, substantially weaker requirement. But LIML-like estimators not robust to heterogeneous treatment effects.
- Solution 2: subtract estimate of bias. Easy to do under homoskedastic errors: leads to variants of the bias-corrected TSLS estimator that dates back to Nagar (1959). But consistency does depend on homoskedasticity.

- Bias caused by using D_i in leniency measure $\hat{\ell}_{\mathrm{TSLS},i}$: obvious solution is to not use it!
- Without covariates, clear we want to use $\hat{\ell}_{\mathrm{JIVE1},i} = Z_i'\hat{\pi}_{-i}$, where $\hat{\pi}_{-i}$ based on regression of D onto Z with i excluded. How to incorporate covariates?
- Option 1 (Angrist, Imbens, and Krueger 1999; Phillips and Hale 1977): construct predictors $\hat{\pi}_{-i}$ and $\hat{\psi}_{D,-i}$ of π and ψ_D based on a regression of D onto (Z, W), but with the ith observation removed, to construct absolute leniency $\hat{\ell}^{abs}_{\mathrm{JIVE1},i} = Z_i'\hat{\pi}_{-i} + W_i'\hat{\psi}_{D,-i}$
    - What is this in judges design?

Jackknife ii

- Then run an IV regression of Y_i onto D_i and W_i, using $\hat{\ell}^{abs}_{\text{JIVE1},i}$ as an instrument for D_i. Resulting estimator known as jackknife instrumental variables estimator (JIVE1):

$$\hat{\beta}_{\text{JIVE1}} = \frac{\ddot{Y}'\hat{\ell}^{abs}_{\text{JIVE1}}}{\ddot{D}'\hat{\ell}^{abs}_{\text{JIVE1}}}, \quad \hat{\ell}^{abs}_{\text{JIVE1}} = \ddot{H}D, \quad \ddot{H} = (I - \operatorname{diag}(H_X))^{-1}(H_X - \operatorname{diag}(H_X)),$$

- Don't need to run n first-stage regressions
- Problem: projecting out covariates from $\hat{\ell}^{abs}_{\text{JIVE1}}$ re-introduces own observation bias
  - In judges design, adjust instrument by average sentencing rate in the location of i, and this average includes own observation
  - Increasing own D_i decreases relative JIVE1 leniency, so expect bias to typically run in opposite direction to TSLS bias.

Jackknife iii

- Under homoskedasticity, JIVE1 bias given by 2SLS bias times

$$-\frac{E[F]L}{(E[F]-1)K-L}.$$

  Can be comparable to or worse than TSLS bias when L large relative to K.

- Leads to option 2: first partial out the covariates, and then do the leave-one-out prediction. Called improved improved jackknife instrumental variables estimator (IJIVE1) estimator, proposed in Ackerberg and Devereux (2009).

$$\hat{\beta}_{\text{IJIVE1}} = \frac{\ddot{D}'\ddot{H}'\ddot{Y}}{\ddot{D}'\ddot{H}'\ddot{D}}, \qquad \ddot{H} = (I - \operatorname{diag}(H_{\ddot{Z}}))^{-1}(H_{\ddot{Z}} - \operatorname{diag}(H_{\ddot{Z}})).$$

Jackknife iv

- Option 3: We want to estimate
$$\beta = \frac{\sum_i E[Y_i \tilde{Z}_i' \pi]}{\sum_i E[D_i \tilde{Z}_i \pi]}.$$
Just use sample analog with $\hat{\pi}_{-i}$ instead of π! That is, use relative leniency $\hat{\ell}_{UJIVE,i} = \ddot{Z}_i' \hat{\pi}_{-i}$ instead of $\hat{\ell}_{\text{TSLS},i} = \ddot{Z}_i' \hat{\pi}$. Kolesár (2013) calls this estimator unbiased jackknife instrumental variables estimator (UJIVE)
    - (Actually, version in the paper slightly inferior, wait for revision. This version analyzed in Kolesár et al. (2026))

  That is, (i) jackknife regression of D onto (W, Z) to compute $\hat{\pi}_{-i}$ (ii) covariate-adjust Z to get $\ddot{Z}$, and then use relative leniency $\hat{\ell}_{\text{UJIVE},i} = \ddot{Z}_i' \hat{\pi}_{-i}$ as single instrument, $\hat{\beta}_{\text{UJIVE}} = \hat{\ell}_{\text{UJIVE}}' Y / \hat{\ell}_{\text{UJIVE}}' D$. Or use `ManyIV` package.

Jackknife v

- Advantage: UJIVE doesn't restrict number of covariates while IJIVE1 consistent so long as $nE[F]/L \to \infty$
- Chao, Swanson, and Woutersen (2023) offer alternative to UJIVE based on Hadamard products (analog to Hadamard variance estimator for high-dimensional OLS), but implementing requires inverting $n \times n$ matrices…Not feasible in e.g., Angrist and Krueger (1991).

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Summary and illustration

- Define $\pi_\Delta = \delta - \pi\beta$, $u_{\Delta,i} = u_{Yi} - u_{Di}\beta$,
  $\gamma = E[W_iW_i']^{-1}E[W_i(Y_i - D_i\beta)] = E[W_iW_i']^{-1}E[W_iZ_i']\pi_\Delta + \psi_Y - \psi_D\beta$, and $\epsilon_i = \tilde{Z}_i'\pi_\Delta + u_{\Delta,i}$.
- Write the "structural" equation as

$$Y_i = D_i\beta + W_i'\gamma + \epsilon_i.$$

- Saw in the previous lecture that the oracle estimator satisfied

$$\mathcal{V}_{1,n}^{-1/2}(\hat{\beta}^* - \beta) \Rightarrow \mathcal{N}(0,1), \qquad \mathcal{V}_{1,n} = \frac{E[\epsilon_i^2(\tilde{Z}_i'\pi)^2]}{E[(\tilde{Z}_i'\pi)^2]^2}.$$

  Variance $\mathcal{V}_{1,n}$ is estimated by the conventional robust standard errors, such as those in Stata. Also correct standard errors for TSLS, UJIVE, or IJIVE1 if (i) there is no treatment effect heterogeneity, and (ii) $E[F] \to \infty$ for UJIVE, or IJIVE1, and $E[F]/K \to \infty$ for TSLS.

More robust variance i

- If (i) fails, then, as discussed last time, don't achieve oracle variance, but instead correct asymptotic variance is given by

$$\mathcal{V}_{2,n} = \frac{E[((\tilde{Z}_i'\pi_\Delta)u_{D,i} + \epsilon_i(\tilde{Z}_i'\pi))^2]}{E[(\tilde{Z}_i'\pi)^2]^2}.$$

- What if (ii) fails?

More robust variance ii

- If $E[F] \not\to \infty$, and use UJIVE or IJIVE1, we need to account for the presence of many instruments in the asymptotic variance formula (Evdokimov and Kolesár 2018, Theorem 5.4)

$$(\mathcal{V}_{2,n} + \mathcal{V}_{MI,n})^{-1/2}(\hat{\beta}_{\text{UJIVE}} - \beta) \Rightarrow \mathcal{N}(0,1),$$
$$\mathcal{V}_{MI,n} = \frac{1}{r_n^2}\sum_{i \neq j}(H_{\tilde{Z},ij}^2 u_{\Delta,i}^2 u_{2,j}^2 + H_{\tilde{Z},ij}^2 u_{\Delta,i}u_{2,i} \cdot u_{\Delta,j}u_{2,j}), \quad (3)$$

- Can show that plug-in variance estimator of $V_{2,n}$ based on UJIVE actually conservative for $\mathcal{V}_{2,n} + \mathcal{V}_{MI,n}$.
- Non-conservative approach requires leave-three-out variance estimator (Kolesár et al. 2026; Yap 2025)

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- Whether to cluster standard errors depends on design. If decision-maker assignment done in batches, then cluster them (e.g. all defendants arriving on Monday assigned same judge). Need to use leave-one-cluster out version of UJIVE (Frandsen, Leslie, and McIntyre 2025; Kolesár et al. 2026). But clustering overused in practice, since this is not typical case!
- Since decision-makers typically quasi-randomly assigned, should start analysis by checking for balance, once necessary covariates W_i included. Easy to do, just run UJIVE with covariate A_i as outcome. Other placebo checks (e.g. using lagged outcomes) can be done the same way.
- Test monotonicity and characterize compliers just as before, except use UJIVE.
- Goldsmith-Pinkham, Hull, and Kolesár (2025) provide a checklist

- Some papers calculate leniency manually, often as a leave-one-out prediction, effectively computing JIVE1 by hand. But this does not mean that $K = 1$! That will overstate actual instrument strength.
- Manually computing leniency runs into other issues as well if code is not maintained as a whole. Don't do it!
- When using the (correctly computed) first-stage F for diagnostics, remember that the $F > 10$ rule of thumb tests hypothesis that TSLS bias, relative to the bias of OLS exceeds 0.1.
- But small F statistic not necessarily a concern when UJIVE or IJIVE1 are used, just need $\sqrt{K}E[F] \to \infty$ for consistency.

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Summary and illustration

- If $E[F]$ small TSLS biased. Instead, use IJIVE1 or UJIVE.
    - $E[F]$ may be small even if K/n small and n large!
    - JIVE1 not a good solution
- To ensure reliable inference, standard errors need to account for additional many instrument term in the asymptotic variance
    - Even more important is to avoid downward bias that's present in the default standard errors estimator based on TSLS—see notes.
- Will now illustrate in 3 applications:
    1. Agan, Doleac, and Harvey (2023), Table III col (3). Outcome is Criminal complaint within two years, treatment is non-prosecution of non-violent misdemeanor defendant. Instrument is prosecutor. Include court-by-month and court-by-day of the week fixed effects. 314 IVs, 1,260 controls

2. Farre-Mensa, Hegde, and Ljungqvist (2020). Treatment is granting of a patent, outcome is whether firm filed a subsequent successful patent application. Instrument is patent examiner. Include industry-by-year fixed effects. 4,238 IVs, 2,401 controls
3. Angrist and Krueger (1991), a relatively tame specification. Outcome is log wages, treatment is years of education, instrument is QOB interacted with year of birth. 30 IVs, and 21 controls (division fixed effects, year fixed effects, black, married, SMSA).

Estimator	Estimate	$\hat{\mathcal{V}}_1^{1/2}$	$\hat{\mathcal{V}}_2^{1/2}$
Panel A: Agan, Doleac, and Harvey (2023). $F = 2.6$, $n = 67,000$.			
OLS	-0.136	0.005	0.005
TSLS	-0.219	0.044	0.053
UJIVE	-0.255	0.069	0.083
IJIVE1	-0.263	0.069	0.083
JIVE1	-0.018	0.050	0.067
Panel B: Farre-Mensa, Hegde, and Ljungqvist (2020). $F = 1.6$, $n = 32,514$.			
OLS	0.223	0.005	0.005
TSLS	0.240	0.012	0.014
UJIVE	0.259	0.041	0.050
IJIVE1	0.250	0.031	0.037
JIVE1	0.111	0.484	0.641
Panel C: Angrist and Krueger (1991). $F = 4.7$, $n = 329,509$.			
OLS	0.063	0.000	0.000
TSLS	0.081	0.016	0.018
UJIVE	0.085	0.021	0.023
IJIVE1	0.085	0.021	0.023
JIVE1	0.090	0.026	0.028

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